Assignment No-1

Page No.
Date

Q.1 Print an $n \times n$ star square.
Take a dynamic input from the user & if print an $n \times n$ square.

ex.
If $n = 4$

```
* * * *
* * * *
* * * *
* * * *
```

```
class P1 {
    public static void main (string [] args){
        ~~for~~
        int n = 4;
        for (int i = 1 ; i ≤ n ; i++) {
            for (int j = 1 ; j ≤ n ; j++) {
                System.out.print ("*");
            }
            System.out.println ();
        } }
    }
```

Dry run :-

(A) $i = 1$ // outer loop $\longrightarrow$ $i \le n$ $(1 \le 4)$ True.

$\longrightarrow$ enter in inner loop.

(1) $j = 1$, $j \le n$ $(1 \le 4)$ True
print ("*"); *
j++;

(2) $j = 2$ $j \le n$ $(2 \le 4)$ * *
print ("*");
j++;

Page No.
Date

③ j = 3 , j ≤ n (3 ≤ 4) True.
print ("*") ; o/p
j++ * * *

④ j = 4 , j ≤ n (4 ≤ 4) True
print ("*") ; o/p
j++ ; * * * *

⑤ j = 5 , j ≤ n (5 ≤ 4) false.
exist loop j

Ⓑ i++;

i = 2 ; i ≤ n ; True // outerloop.
enter in loop j // innerloop

j = 1 ; j ≤ n (1 ≤ 4) True
print (*) ; o/p
j++; * * * *
*

j = 2 ; j ≤ n (2 ≤ 4) True.
print (*) ; op * * * *
j++; * *

j = 3 ; j ≤ n (3 ≤ 4) True
print (*) ; op
j++ ; * * * *
* * *

j = 4 j ≤ n (4 ≤ 4) True
print (*) ; o/p
j++ ; * * * *
* * * *

j = 5 ; j ≤ n (5 ≤ 4)
false exist.

i++

i = 3 ; i ≤ n (3 ≤ 4) True.

j = 1 j ≤ n (1 ≤ 4) True

print (*) & j++ o/p * * * *

* * * *

*

j = 2 j ≤ n (2 ≤ 4) True.

print (*) & j++ ; (3,1) (3,2) (3,3) (3,4)

* *

j = 3 ; j ≤ n (3 ≤ 4) True.

print (*) & j++ (3,1) (3,2) (3,3) (3,4)

* * *

j = 4 ; j ≤ n (4 ≤ 4) True

print (*) & j++ (3,1) (3,2) (3,3) (3,4)

* * * *

j = 5 j ≤ n (5 ≤ 5) false

exit

→ i++ ;

i = 4 i ≤ n (4 ≤ 4) True.

j = 1 ; j ≤ n (1 ≤ n) True o/p

print (*) & j++ ; (4,1) (4,2) (4,3) (4,4)

*

j = 2 ; j ≤ n (2 ≤ 4) True

print (*) & j++ (4,1) (4,2) (4,3) (4,4)

* *

j = 3 ; j ≤ n (3 ≤ 4) True

print (*) & j++ (4,1) (4,2) (4,3) (4,4)

* * *

j = 4 ; j ≤ n (4 ≤ 4) True.

print (*) & j++ (4,1) (4,2) (4,3) (4,4

* * * *

j = 5 j ≤ n (5 ≤ 4) false exit

i++ ; i = 5 (i ≤ n) 5 ≤ 4 false.

Page No.
Date

Q. 2 print the following pattern.

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

code.

```
n=4
for (int i=1 ; i≤n ; i++){
    for (int j=1 ; j≤n ; j++){
        sop (i);
    }
    sop ();
}
```

Dry run.

~~code~~

i=1 i≤n (1≤4) →

j=1 j≤n (1≤4) True
print (i) & j++

j=2 ; j≤n (2≤4) True
print (i) & j++

j=3 ; j≤n (3≤4) True.
print (i) & j++

j=4 j≤n (4≤4) True
print (i) & j++

j=5 j≤n (5≤4) false
બહાર loop ના બહાર

Page No.
Date

i++; i = 2 i ≤ n (2 ≤ 4) True

↓

j = 1 ; j ≤ n (1 ≤ 4) True
print (i) & j++;

j = 2 , j ≤ n (2 ≤ 4) True
print (i) & j++

j = 3 j ≤ n (3 ≤ 4) True
print (i) & j++

j = 4 ; j ≤ n (4 ≤ 4) True
print (i) & j++

j = 5 ; j ≤ n (5 ≤ 4) false
j loop બહાર નીકળે.

i++ ; i = 3 i ≤ n (3 ≤ 4) True

↓

j = 1 j ≤ n (1 ≤ 4) True
print (i) & j++

j = 2 j ≤ n (2 ≤ 4) True
print (i) & j++

j = 3 j ≤ n (3 ≤ 4) True
print (i) & j++

j = 4 j ≤ n (4 ≤ 4) True
print (i) & j++

j = 5 j ≤ n (5 ≤ 4) false
j loop બહાર નીકળે.

i++

i = 4 ; i ≤ n (4 ≤ 4) True.

↓

j = 1 j ≤ n (1 ≤ 4) True.

print (i) & j++

j = 2 j ≤ n (2 ≤ 4) True.

print (i) & j++

j = 3 3 ≤ 4 True.

print (i) & j++

j = 4 4 ≤ 4 True.

print (i) & j++

j = 5 5 ≤ 5 false.

j loop से बाहर.

i++

i = 5 5 ≤ 4 false.

i loop से बाहर

final o/p

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

Page No.
Date

Q.3 print the following pattern

n = 4

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

code.

```
int n = 4
for (int i = 1 ; i ≤ n ; i++) {
    for (int j = 1 ; j ≤ n ; j++) {
        sop (j) ;
    }
    sop () ;
}
```

cond^n

→ $i = 1$ $i \le n$ $(1 \le n)$ True.

$j = 1$ $j \le n$ $(1 \le n)$ → print (j) → j++

$j = 2$ $2 < 4$ → print (j) → j++

$j = 3$ $3 < 4$ → print (j) → j++

$j = 4$ $4 \le 4$ → print (j) → j++

$j = 5$ $5 \le 5$ ✗ cond^n false

i++ ; $i = 2$ $i \le n$ $(2 \le n)$ True.

$j = 1$ $1 \le 4$ → print (j) → j++

$j = 2$ $2 \le 4$ → print (j) → j++

$j = 3$ $3 \le 4$ → print (j) → j++

Page No.
Date

$j = 4$ $\quad 4 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

~~j++~~

$j = 5$ $\quad 5 \leq 4$ false.

i++ , $i = 3$ $\quad 3 \leq 4$ True.

$j = 1$ $\quad 1 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

$j = 2$ $\quad 2 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

$j = 3$ $\quad 3 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

$j = 4$ $\quad 4 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

$j = 5$ $\quad 5 \leq 4$ false.

i++ $\quad i = 4$ $\quad 4 \leq 4$ True.

$j = 1$ $\quad 1 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++
$j = 2$ $\quad 2 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++
$j = 3$ $\quad 3 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++
$j = 4$ $\quad 4 \leq 4 \longrightarrow$ print(j) $\longrightarrow$ j++

$j = 5$ $\quad 5 \leq 4$ false.

i++ $\quad i = 5$ $\quad 5 \leq 4$ false

final o/p

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4